

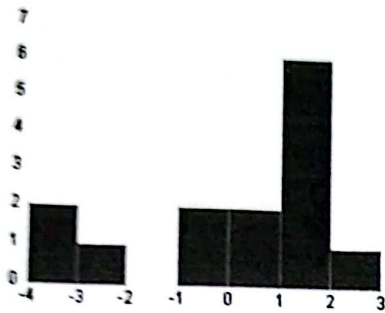
1. Fie seria formata din investitiile straine (FDI) in perioada 2007-2021 pentru Danemarca (3 p).
- a) Sa se analizeze stationaritatea seriei pe baza corelogramei.
- b) Sa se identifice tipul de proces urmat pe baza outputului din Eviews pentru seria integrate de ordin 1.
- c) Precizati tipul de proces urmat de seria analizata si daca acesta poate fi validat.

Correlogram of FDI						Correlogram of D(FDI)					
Date: 06/05/24 Time: 13:13 Sample (adjusted): 2007 2020 Included observations: 14 after adjustments						Date: 06/05/24 Time: 13:14 Sample (adjusted): 2008 2020 Included observations: 13 after adjustments					
Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob	Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
		1 -0.418	-0.418	3.0174	0.082			1 -0.745	-0.745	9.0203	0.003
		2 0.289	0.138	4.5774	0.101			2 0.425	-0.292	12.225	0.002
		3 -0.167	-0.006	5.1451	0.161			3 -0.142	0.123	12.617	0.006
		4 -0.050	-0.188	5.2007	0.267			4 -0.010	0.059	12.620	0.013
		5 -0.192	-0.308	6.1166	0.295			5 0.081	0.043	12.781	0.026
		6 -0.213	-0.471	7.3900	0.286			6 -0.240	-0.375	14.386	0.026
		7 0.227	0.026	9.0318	0.250			7 0.273	-0.228	16.802	0.019
		8 -0.149	0.035	9.8652	0.275			8 -0.250	-0.043	19.232	0.014
		9 0.163	-0.129	11.061	0.272			9 0.151	0.022	20.340	0.016
		10 0.015	-0.179	11.074	0.352			10 -0.069	-0.029	20.649	0.024
		11 0.067	-0.188	11.408	0.410			11 0.048	-0.020	20.875	0.035
		12 -0.064	-0.156	11.868	0.456			12 -0.023	-0.095	20.975	0.051

Dependent Variable: FDI
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 06/05/24 Time: 13:15
Sample: 2007 2020
Included observations: 14
Convergence achieved after 19 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.569997	0.664801	0.857394	0.4113
AR(1)	-0.621420	0.608040	-1.022005	0.3309
MA(1)	0.232328	0.885210	0.262455	0.7983
SIGMASQ	4.658983	3.510291	1.327236	0.2139

R-squared	0.208646	Mean dependent var	0.631303
Adjusted R-squared	-0.028760	S.D. dependent var	2.517981
S.E. of regression	2.553933	Akaike info criterion	4.964651
Sum squared resid	65.22576	Schwarz criterion	5.147238
Log likelihood	-30.75255	Hannan-Quinn criter.	4.947749
F-statistic	0.878857	Durbin-Watson stat	1.806400
Prob(F-statistic)	0.484359		



Series: Residuals	
Sample 2007 2020	
Observations 14	
Mean	0.057181
Median	0.800911
Maximum	2.800753
Minimum	-3.921490
Std. Dev.	2.230162
Skewness	-0.752199
Kurtosis	2.100300
Jarque-Bera	1.000039
Probability	0.4280700

Heteroskedasticity Test: ARCH

F-statistic	0.016146	Prob. F(1,11)	0.9012
Obs*R-squared	0.019053	Prob. Chi-Square(1)	0.8902

Test Equation:
 Dependent Variable: RESID*2
 Method: Least Squares
 Date: 06/05/24 Time: 13:17
 Sample (adjusted): 2008 2020
 Included observations: 13 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	4.604711	2.132906	2.158891	0.0538
RESID*2(-1)	-0.038542	0.303323	-0.127066	0.9012
R-squared	0.001466	Mean dependent var	4.413964	
Adjusted R-squared	-0.089310	S.D. dependent var	5.234328	
S.E. of regression	5.463067	Akaike info criterion	6.374536	
Sum squared resid	328.2962	Schwarz criterion	6.461451	
Log likelihood	-39.43448	Hannan-Quinn criter.	6.356671	
F-statistic	0.016146	Durbin-Watson stat	1.878016	
Prob(F-statistic)	0.901182			

2. Pentru seria prezentata la punctul 1, testele de stationaritate se prezinta astfel (3 p):
 - a) Specificati daca seria este stationara.
 - b) Identificati testele folosite si scrieti ipotezele si regula de decizie.
 - c) Identificati tipul de ecuatie folosit.
 - d) Sugerati o modalitate prin care seria aceasta ar putea deveni stationara.

KPSS Unit Root Test on FDI	
Null Hypothesis: FDI is stationary	
Exogenous: Constant, Linear Trend	
Bandwidth: 1 (Newey-West automatic) using Bartlett kernel	
LM-Stat	
Kwiatkowski-Phillips-Schmidt-Shin test statistic	0.112728
Asymptotic critical values*: 1% level	0.216000
5% level	0.146000
10% level	0.119000
*Kwiatkowski-Phillips-Schmidt-Shin (1992, Table 1)	
Residual variance (no correction)	5.885262
HAC corrected variance (Bartlett kernel)	3.438807

KPSS Unit Root Test on D(FDI)	
Null Hypothesis: D(FDI) is stationary	
Exogenous: Constant, Linear Trend	
Bandwidth: 1 (Newey-West automatic) using Bartlett kernel	
LM-Stat	
Kwiatkowski-Phillips-Schmidt-Shin test statistic	0.130515
Asymptotic critical values*: 1% level	0.216000
5% level	0.146000
10% level	0.119000
*Kwiatkowski-Phillips-Schmidt-Shin (1992, Table 1)	
Residual variance (no correction)	17.02692
HAC corrected variance (Bartlett kernel)	4.277100

3. Pentru a testa existența unei corelații semnificative Investițiile straine (FDI) și Indicele de dezvoltare umană (HDI), s-au folosit date anuale, rezultatele fiind prezentate mai jos (3 p):
 - a) Există cauzalitate între cele două variabile?
 - b) Există o relație de cointegrare între seriile analizate? Definiți și exemplificați conceptul de cointegrare.
 - c) Interpretați economic rezultatele din outputul (VAR) și cuantificați impactul HDI-ului asupra emisiilor de CO₂.

d) Justificati daca aplicarea modelului VECM ar conduce la rezultate mai bune.

Pairwise Granger Causality Tests

Date: 06/05/24 Time: 13:19

Sample: 2007 2021

Lags: 2

Null Hypothesis:	Obs	F-Statistic	Prob.
HDI does not Granger Cause FDI	12	0.48261	0.6363
FDI does not Granger Cause HDI		0.31287	0.7411

Johansen Cointegration Test

Date: 06/05/24 Time: 13:20

Sample (adjusted): 2010 2020

Included observations: 11 after adjustments

Trend assumption: Linear deterministic trend

Series: DFDI DHDI

Lags interval (in first differences): 1 to 1

Unrestricted Cointegration Rank Test (Trace)

Hypothesized No. of CE(s)	Eigenvalue	Trace Statistic	0.05 Critical Value	Prob.**
None *	0.798123	27.89077	15.49471	0.0004
At most 1 *	0.607582	10.28972	3.841465	0.0013

Vector Autoregression Estimates

Date: 06/05/24 Time: 13:21

Sample (adjusted): 2010 2020

Included observations: 11 after adjustments

Standard errors in () & t-statistics in []

	DFDI	DHDI
DFDI(-1)	-1.033241 (0.37173) [-2.77957]	-0.000146 (0.00040) [-0.36079]
DFDI(-2)	-0.312482 (0.37490) [-0.83350]	0.000100 (0.00041) [0.24580]
DHDI(-1)	82.28285 (221.426) [0.37160]	0.126543 (0.24067) [0.52580]
DHDI(-2)	-132.7210 (221.967) [-0.59793]	-0.568372 (0.24126) [-2.35587]
C	-0.120416 (1.36665) [-0.08811]	0.004688 (0.00149) [3.15620]
R-squared	0.667903	0.506915
Adj R-squared	0.446504	0.178192